Lecture 1

Course Content

- **Introduction to Cloud Computing** (Definition, Evolution, Characteristics, Benefits)
- **Cloud Service Models** (SaaS, PaaS, IaaS, DaaS, FaaS)
- **Cloud Deployment Models** (Public, Private, Hybrid)
- **Cloud Computing Architecture** (Virtualization, Datacenters, Virtual Machines)
- **Virtualization Technology** (Type 1 and Type 2, VMs and Containers, Full & Para)
- **Cloud Storage** (Object, File & Block Storage), Providers, Backups, Archives
- **Cloud Security and Privacy** (Privacy and Compliance, IAM, ISO 27001, NIST)
- **Cloud Networking** (VPC, SDN, Load Balancing)
- **Cloud Application Development** (DevOps, CI/CD, Containerization & Kubernetes)
- **Current Trends in Cloud Computing**
- **Cloud Platforms & Tools** (AWS – EC2, S3, RDS, Lambda; Azure – ARM; GCP, etc.)

Slide 1: Introduction to Cloud Computing

- Cloud computing is the delivery of computing services over the internet
- Services include servers, storage, databases, networking, software
- Resources are provided on-demand
- Pay-as-you-go pricing model
- No need for physical infrastructure

Slide 2: Evolution of Cloud Computing

- Mainframe computing
- Client–server computing
- Distributed computing
- Grid computing

- Cloud computing

Slide 3: Characteristics of Cloud Computing

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity and scalability
- Measured service

Slide 4: Benefits of Cloud Computing

- Cost efficiency
- Scalability and flexibility
- High availability
- Disaster recovery
- Reduced maintenance

Slide 5: Cloud Service Models

- Software as a Service (SaaS)
- Platform as a Service (PaaS)
- Infrastructure as a Service (IaaS)
- Desktop as a Service (DaaS)
- Function as a Service (FaaS)

Slide 6: SaaS, PaaS, IaaS

- **SaaS**: Ready-to-use applications
- **PaaS**: Platform for application development

- **IaaS**: Virtualized computing resources
- Users access services via internet
- Provider manages infrastructure

Slide 7: Cloud Deployment Models

- Public Cloud
- Private Cloud
- Hybrid Cloud

Slide 8: Public, Private, Hybrid Cloud

- **Public Cloud**: Shared resources, cost-effective
- **Private Cloud**: Dedicated resources, high security
- **Hybrid Cloud**: Combination of public and private

Slide 9: Cloud Computing Architecture

- Front-end (client side)
- Back-end (cloud side)
- Virtualization layer
- Datacenters
- Network connectivity

Slide 10: Virtualization Technology

- Virtualization enables multiple OS on one hardware
- Improves resource utilization
- Reduces hardware cost
- Enables isolation

Slide 11: Hypervisors

- Type 1 (Bare-metal hypervisor)
- Type 2 (Hosted hypervisor)
- Examples: VMware, Hyper-V, VirtualBox

Slide 12: VMs and Containers

- Virtual Machines run full OS
- Containers share host OS
- Containers are lightweight
- Faster deployment using containers

Slide 13: Types of Virtualization

- Full virtualization
- Paravirtualization
- Hardware-assisted virtualization

Slide 14: Cloud Storage

- Object storage
- File storage
- Block storage
- Scalable and durable

Slide 15: Cloud Storage Management

- Cloud storage providers
- Data backups

- Data recovery
- Archival storage

Slide 16: Cloud Security and Privacy

- Data confidentiality
- Data integrity
- Data availability
- Shared responsibility model

Slide 17: Security Standards and Compliance

- Identity and Access Management (IAM)
- ISO 27001
- NIST standards
- Regulatory compliance

Slide 18: Cloud Networking

- Virtual Private Cloud (VPC)
- Software Defined Networking (SDN)
- Load balancing
- Network security

Slide 19: Cloud Application Development

- DevOps practices
- Continuous Integration (CI)
- Continuous Deployment (CD)
- Automation tools

Slide 20: Containerization and Kubernetes

- Application packaging using containers
- Container orchestration
- Kubernetes for scaling and management
- High availability

Slide 21: Current Trends in Cloud Computing

- Serverless computing
- Edge computing
- Multi-cloud strategy
- AI and ML integration

Slide 22: Cloud Platforms & Tools

- **AWS**: EC2, S3, RDS, Lambda
- **Microsoft Azure**: ARM
- **Google Cloud Platform (GCP)**
- Popular cloud providers

Introduction to Cloud Computing

Cloud computing is a technology that allows users to access computing resources—such as **servers, storage, databases, software, and networking—over the Internet** instead of owning and maintaining physical hardware. These resources are provided by cloud service providers on a **pay-as-you-use** basis and can be accessed anytime, from anywhere.

Instead of installing software or storing data on a personal computer, users can use online services hosted in large **data centers**. Cloud computing offers key features like **on-demand access, scalability, flexibility, cost efficiency, and high availability**.

Example

A common example of cloud computing is **Google Drive**:

- Files are stored on Google's cloud servers, not on your local computer.
- You can access your documents from a laptop, mobile phone, or tablet.
- No need to buy extra storage hardware.
- Data is automatically backed up and synchronized.

Other examples include:

- **Gmail** (email service)
- **Amazon Web Services (AWS)** for hosting websites
- **Microsoft Azure** for application development
- **Netflix**, which streams content using cloud infrastructure

What is Cloud Computing?

Cloud computing means delivering **applications and IT infrastructure over the Internet** instead of running them on local computers or servers.

1. Cloud Computing: Apps and Infrastructure over the Internet

- Software, storage, servers, and networks are accessed online
- No need to install or maintain heavy hardware locally

Example: Gmail, Google Drive, Microsoft Office 365

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2. Compute as a Service: Applications over the Internet

- Computing power and applications are provided as a service
- Users just use the application; infrastructure is managed by the provider

Example: Using AWS EC2 to run applications without owning servers

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3. Utility Computing (Pay-As-You-Go)

- Similar to electricity or water billing
- You pay only for what you use (CPU, storage, bandwidth)

Example: Paying only for the hours a virtual server runs on AWS or Azure

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Three New Aspects of Cloud Computing

1. Illusion of Infinite Computing Resources

- Resources appear unlimited
- Users can scale up or down anytime

Example: A website handling sudden traffic during sales

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2. No Upfront Commitment

- No need to buy expensive hardware in advance
- Start small and expand when needed

Example: A startup launching an app without buying servers

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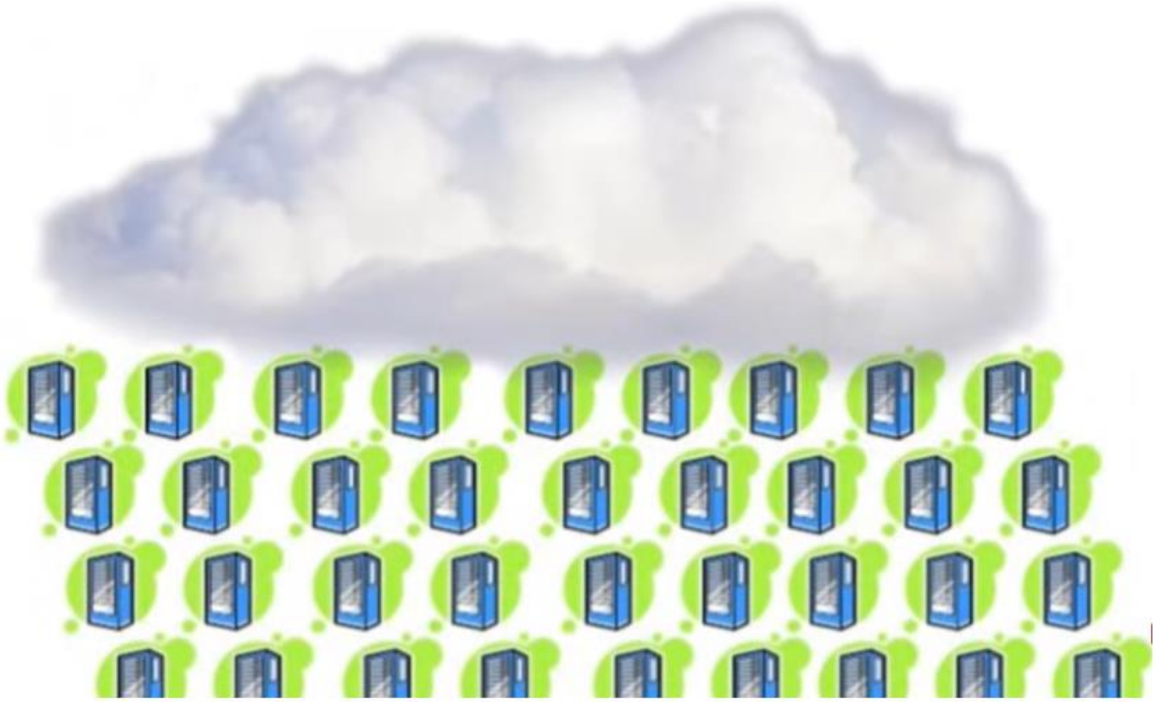
3. Pay for Short-Term Usage

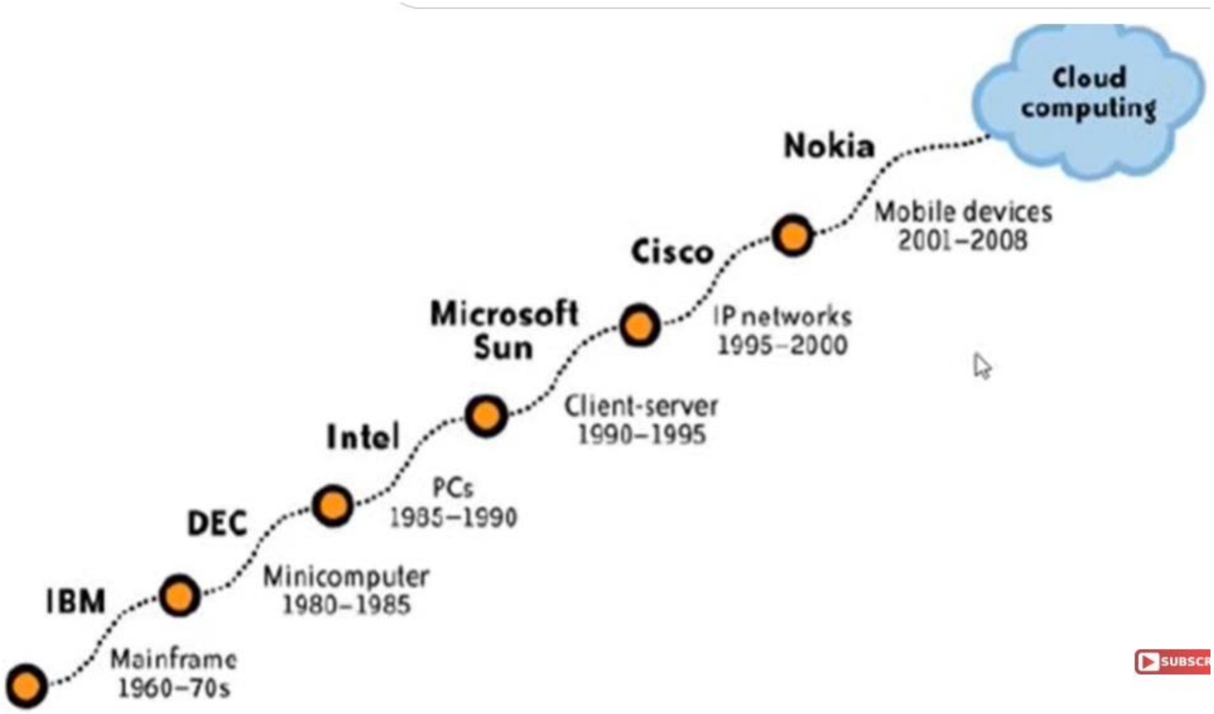
- Resources can be used temporarily
- Billed per hour, minute, or second

Example: Using cloud servers only during exams or peak business hours

and very different from a PC...



Evolution of Computing (Explanation)

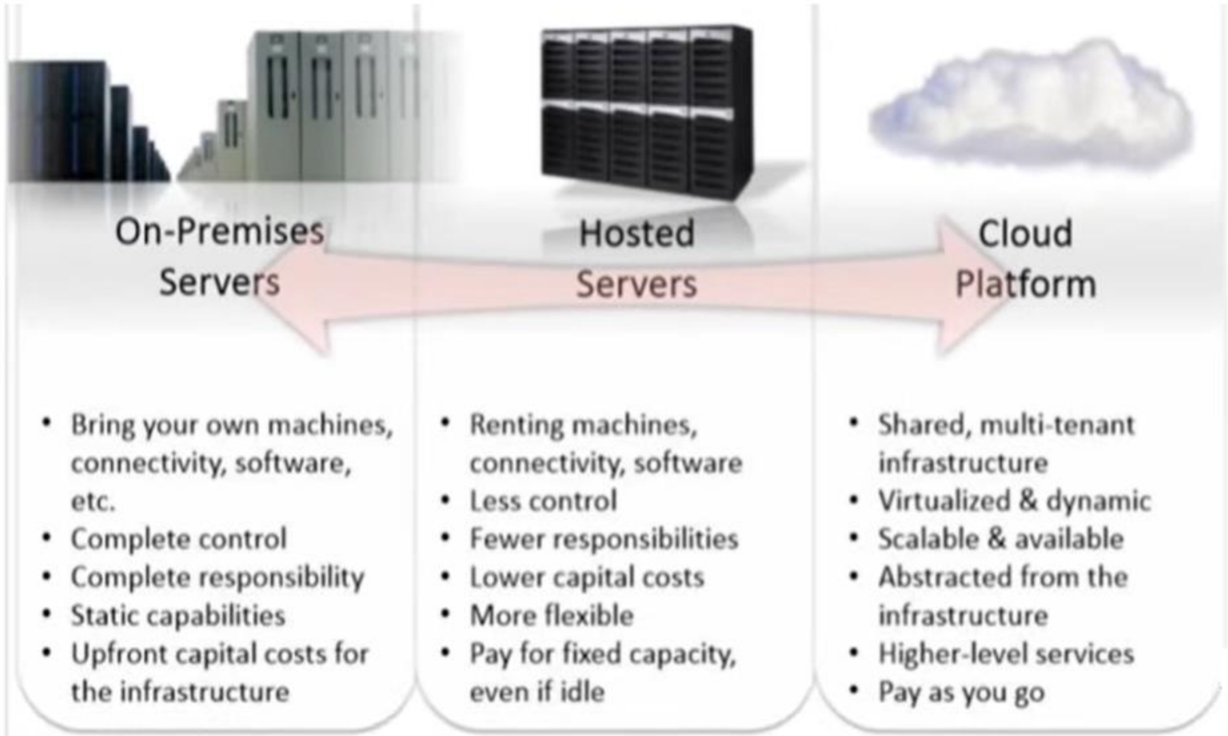


1. **Mainframe Computers (1960s–1970s) – IBM**
    - Very large and expensive computers
    - Used by governments and big organizations
    - All processing was centralized
    - Users accessed systems through terminals
2. **Minicomputers (1980–1985) – DEC**
    - Smaller and cheaper than mainframes
    - Used by universities and medium-sized companies
    - Supported multiple users at the same time
3. **Personal Computers (1985–1990) – Intel**
    - Computers became affordable for individuals
    - Standalone systems used at homes and offices

- Each user had their own machine

4. **Client–Server Computing (1990–1995) – Microsoft, Sun**
   - Servers handled data and processing
   - Clients (PCs) requested services from servers
   - Popular in enterprise networks
5. **IP Networks / Internet Era (1995–2000) – Cisco**
   - Growth of the Internet and networking
   - Data and services became globally accessible
   - Enabled web-based applications
6. **Mobile Computing (2001–2008) – Nokia**
   - Rise of mobile phones and portable devices
   - Users accessed services anytime, anywhere
   - Increased demand for online storage and apps
7. **Cloud Computing (Present)**
   - Computing resources delivered over the Internet
   - On-demand access to servers, storage, and applications
   - Scalable, cost-effective, and flexible
   - Examples: AWS, Azure, Google Cloud

IT infrastructure



1. On-Premises Servers

(Traditional setup – everything in your own office)

Explanation:

- The organization **buys and owns** all hardware and software.
- Servers are installed and managed **locally**.
- Full control, but also **full responsibility**.

Key Points:

- Bring your own machines, connectivity, and software
- Complete control and responsibility
- Fixed (static) capacity
- High upfront capital cost
- Maintenance, security, and upgrades are your job

Example:
A university maintains its own data center for student records and exam systems.

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2. Hosted Servers

(Renting servers from a provider)

Explanation:

- Servers are hosted in a **third-party data center**.
- You rent the hardware instead of buying it.
- Some management is handled by the provider.

Key Points:

- Renting machines, software, and connectivity
- Less control than on-premises
- Fewer responsibilities
- Lower initial cost
- More flexible than on-premises
- Pay for fixed capacity, even when idle

Example:
A company rents dedicated servers from a hosting provider to run its website.

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3. Cloud Platform

(Modern cloud computing model)

Explanation:

- Computing resources are provided **on-demand over the internet**.
- Infrastructure is shared and highly virtualized.
- You only pay for what you use.

Key Points:

- Shared, multi-tenant infrastructure
- Virtualized and dynamic resources
- Highly scalable and always available
- Infrastructure is abstracted (you don't manage hardware)
- Higher-level services (databases, analytics, AI)
- Pay-as-you-go pricing

Example:
Using **AWS EC2** for servers and **S3** for storage, scaling up during peak traffic and scaling down later.

Overall Comparison

Feature	On-Premises	Hosted	Cloud
Control	Very High	Medium	Low
Cost	High upfront	Medium	Pay-as-you-use
Scalability	Low	Limited	Very High
Responsibility	Full	Shared	Minimal